A rank obstruction in prime-degree Suzuki groups

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Abstract

We prove the nonexistence assertion in Theorem 5.4 of the fixed 1999 rank-three Suzuki paper. The argument uses its subgroup definition of residual weak primitivity directly, so assumes neither flag transitivity nor a geometric interpretation of every residue. Two successive maximal intersections suffice. The required proper-overgroup information is taken from the classical Suzuki subgroup classification, with the elementary point-stabilizer calculation imported from the separately reviewed note.

1 The subgroup condition

Let I be finite and let $(G_i)_{i\in I}$ be subgroups of a finite group G. Write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The condition used in the source [1], printed page 46, is

$$\text{for every } J \subsetneq I \text{ there is } i \in I \setminus J \text{ such that } G_{J\cup\{i\}} \text{ is maximal proper in } G_J. \tag{1}$$

Here $J = \varnothing$ is included. The native statement of Theorem 5.4, printed page 55, excludes rank *at least* three, not merely rank strictly greater than three.

Lemma 1. *Under* (1)*, for every $J \subsetneq I$ and every $i \notin J$, $G_{J\cup\{i\}} < G_J$. In particular all G_i are proper, no G_i is contained in a different G_j, and adjoining any further type to an intersection strictly decreases it.*

Proof. If $G_J \le G_i$, then $G_{I\setminus\{i\}} \le G_J \le G_i$. Consequently $G_{I\setminus\{i\}} = G_I$, contradicting (1) at $I \setminus \{i\}$, where the only available type is i. This proves the asserted strictness and its consequences. □

2 Classical inputs and the point stabilizer

Put $q = 2^n$, with $n \ge 3$ odd, and $G = \mathrm{Sz}(q)$. We retain the classical Suzuki subgroup and ovoid structure used in the source and specified in the accepted notes [2, 3]. The maximal subgroups have types

$$B = P \rtimes C_{q-1}, \quad D = C_{q-1} \rtimes C_2, \quad N_\epsilon = C_{a_\epsilon} \rtimes C_4, \quad \mathrm{Sz}(2^d), \qquad a_\epsilon = q + \epsilon 2^{(n+1)/2} + 1,$$

where the last type has $d \ge 3$ and n/d prime. Every proper subgroup is contained in one of these groups. The three odd torus orders are pairwise coprime. Every nonidentity odd element is in a unique cyclic torus; its centralizer is that torus. A split torus has normalizer D, whose involution inverts it. In N_ϵ the order-four complement acts without nonidentity fixed points on its odd kernel, and its square acts by inversion. The group P has order q^2 and centre Z elementary abelian of order q. A nonidentity involution fixes exactly one ovoid point; B is a point stabilizer. These are attributed structural premises, not new classification claims.

The reviewed elementary proof in [2] shows that every maximal subgroup of B contains Z. We record its consequence for (1), separately from any flag-transitivity hypothesis.

Lemma 2. *If $|I| \geq 2$ and (1) holds, no G_i is a point stabilizer B.*

Proof. Suppose $G_0 = B$. Choose $i \neq 0$ such that $J = B \cap G_i$ is maximal in B. Then $Z \leq J$. No dihedral or nonsplit torus normalizer contains Z: their Sylow two-subgroups have order two or are cyclic of order four. Nor does a proper simple subfield group $\mathrm{Sz}(s)$ contain Z, since $q \geq s^3 > s^2$, the order of its Sylow two-subgroups. A point stabilizer containing Z must equal B, because any fixed nonidentity involution in Z has a unique fixed ovoid point. Thus B is the only maximal subgroup of G containing J. The proper subgroup G_i lies in a maximal subgroup, so $G_i \leq B$, contrary to Lemma 1. □

3 Two maximal intersections

For the remainder assume n is prime. There are then no proper simple subfield groups. This hypothesis is essential to the following proper-overgroup arguments.

Lemma 3. *If x has nontrivial odd order dividing a_ϵ, every proper subgroup containing x lies in the unique N_ϵ associated with its torus. If a dihedral subgroup E has a nontrivial odd rotation subgroup in a split torus, every proper subgroup containing E lies in the normalizer D of that split torus.*

Proof. For the first assertion, take a maximal subgroup containing the given proper subgroup. Coprimality excludes point stabilizers, split normalizers and the opposite nonsplit type. In a nonsplit normalizer an odd-order element belongs to its odd kernel. Uniqueness of the torus containing x therefore identifies that maximal subgroup.

For the second assertion, coprimality excludes both nonsplit types. A point stabilizer cannot contain E. Indeed, its quotient by its normal two-subgroup is abelian of odd order. An odd rotation has nontrivial image in this quotient, whereas the involution inverting it has trivial image. The inversion relation would force that odd image to equal its inverse, a contradiction. The remaining maximal subgroup is a split normalizer, and the rotation subgroup determines its unique torus, hence determines this normalizer. □

Lemma 4. *If n is prime and $|I| \geq 3$, no G_i is a full nonsplit torus normalizer $N = C_a \rtimes C_4$.*

Proof. Suppose $G_0 = N$. For $i \neq 0$, the subgroup $K_i = G_0 \cap G_i$ cannot contain a nonidentity odd-order element: Lemma 3 would give $G_i \leq N$, violating Lemma 1. Thus K_i is a two-group, cyclic of order at most four. Choose i with K_i maximal in N. It cannot have order one or two, since it lies in a Sylow two-subgroup of order four. Hence $K_i \cong C_4$.

Apply (1) at $\{0, i\}$ to choose $k \notin \{0, i\}$ such that G_{0ik} is maximal in K_i; it has order two. An involution $z \in N$ lies in a unique cyclic subgroup of order four. To see this, conjugate its Sylow subgroup to the specified complement V. Then z is its involution and $\mathrm{C}_N(z) = V$, because z inverts the odd kernel. Any cyclic group of order four containing z centralizes z and therefore equals V. It follows that the cyclic two-subgroup K_k containing G_{0ik} is contained in K_i. Hence $G_{0ik} = G_{0k}$, again contradicting Lemma 1. □

Lemma 5. *Suppose n is prime and $q - 1$ is composite. If $|I| \geq 3$, no G_i is the full split normalizer $D = C_{q-1} \rtimes C_2$.*

Proof. Suppose $G_0 = D$, and write $A = C_{q-1}$. Each $K_i = G_0 \cap G_i$, $i \neq 0$, is either contained in A or is a subgroup of order at most two. Indeed, any other subgroup of the odd dihedral group D is dihedral with a nontrivial odd rotation subgroup; Lemma 3 would force $G_i \leq D$. Choose i such that K_i is maximal in D. The maximal subgroups of D are A and dihedral groups of order $2(q-1)/p$, where p is a prime divisor of $q - 1$. The latter have nontrivial odd rotation subgroups because $q - 1$ is composite, so $K_i = A$. Now choose $k \notin \{0, i\}$ with G_{0ik}

maximal in A. Its order is $(q-1)/p > 1$ for some prime p, so is odd and nontrivial. The preceding restriction on K_k forces $K_k \leq A = K_i$. Thus $G_{0ik} = G_{0k}$, contrary to Lemma 1. □

Theorem 6. *Let $n \geq 3$ be an odd prime and suppose $2^n - 1$ is composite. There is no subgroup family for $G = \mathrm{Sz}(2^n)$ satisfying (1) with $|I| \geq 3$. In particular there is no residually weakly primitive pre-geometry of rank at least three in the source's sense.*

Proof. The case $J = \varnothing$ in (1) makes at least one G_i maximal in G. The classical list leaves B, D, N_+, N_-, since n is prime. Lemmas 2, 4 and 5 exclude all four possibilities. □

This proves exactly the source's Theorem 5.4, including its full rank range. The point-stabilizer exclusion also proves the subgroup form of its quoted Lemma 4.1. The nonsplit and split arguments above are only claimed in prime field degree; they do not by themselves dispose of all composite-degree assertions in Lemmas 4.2–4.6. No sufficient realization has been used as a necessity statement. No finite group search, new conjecture resolution, Lean verification or compression measurement is claimed for this component.

References

[1] D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Fixed twenty-one-page edition, especially printed pp. 46 and 55.

[2] Human Compression project, *Maximal subgroups and elementary reductions for Suzuki geometries*, accepted preliminary component for L115, with its explicit classical Suzuki premises.

[3] Human Compression project, *Normalizer fibres in rank-two Suzuki geometries*, accepted L115 component. Its structural inputs are attributed there to Suzuki, Tits and Lüneburg.